

Supplementary Materials: Excess Risk Bounds, Inversion Conditioning, and Calibration Distortion under Class-Conditional Label Noise

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1 Detailed Mathematical Proofs

1.1 Lemma: Matrix Inversion Perturbation via Neumann Series

Lemma 1. Let $T, \hat{T} \in \mathbb{R}^{K \times K}$ be non-singular matrices, and let $E = \hat{T} - T$ denote the estimation error with $\|E\|_2 \leq \|E\|_F \leq \epsilon < \frac{1}{\|T^{-1}\|_2}$. Then:

$$\|\hat{T}^{-1} - T^{-1}\|_2 \leq \frac{\|T^{-1}\|_2^2 \epsilon}{1 - \|T^{-1}\|_2 \epsilon}. \quad (1)$$

Proof. We write $\hat{T} = T + E = T(I + T^{-1}E)$. Since $\|T^{-1}E\|_2 \leq \|T^{-1}\|_2 \|E\|_2 \leq \|T^{-1}\|_2 \epsilon < 1$, the matrix $I + T^{-1}E$ is invertible by the Neumann series theorem, with:

$$(I + T^{-1}E)^{-1} = \sum_{k=0}^{\infty} (-T^{-1}E)^k = I - T^{-1}E + (T^{-1}E)^2 - \dots \quad (2)$$

Consequently:

$$\hat{T}^{-1} = (I + T^{-1}E)^{-1}T^{-1} = \sum_{k=0}^{\infty} (-T^{-1}E)^k T^{-1}. \quad (3)$$

Subtracting T^{-1} yields:

$$\hat{T}^{-1} - T^{-1} = \sum_{k=1}^{\infty} (-T^{-1}E)^k T^{-1} = -(I + T^{-1}E)^{-1} T^{-1} E T^{-1}. \quad (4)$$

Taking operator norms:

$$\|\hat{T}^{-1} - T^{-1}\|_2 \leq \|(I + T^{-1}E)^{-1}\|_2 \|T^{-1}\|_2^2 \|E\|_2 \leq \frac{\|T^{-1}\|_2^2 \|E\|_2}{1 - \|T^{-1}E\|_2} \leq \frac{\|T^{-1}\|_2^2 \epsilon}{1 - \|T^{-1}\|_2 \epsilon}. \quad (5)$$

□

1.2 Proposition 2A: Pointwise Expected Risk Bias

Proposition 1. Let T be an invertible row-stochastic transition matrix, and let $\hat{T}$ satisfy $\|E\|_F \leq \epsilon < 1/\|T^{-1}\|_2$. Assume $0 \leq \ell \leq M$. Then for any measurable $f: \mathcal{X} \rightarrow \Delta^{K-1}$:

$$\left| \mathbb{E}_{\hat{\mathcal{D}}}[\hat{T}^{-1} \vec{\ell}(f(X))]_{\hat{Y}} - R_{\mathcal{D}}(f) \right| \leq \frac{\sqrt{K}M \|T^{-1}\|_2^2 \epsilon}{1 - \|T^{-1}\|_2 \epsilon}. \quad (6)$$

Proof. Let $\tilde{\mathbf{p}}(x) \in \Delta^{K-1}$ be the corrupted class posterior row vector at x . By total probability, $\tilde{\mathbf{p}}(x) = \mathbf{p}(x)T$. By Proposition 1, the clean risk satisfies $R_{\mathcal{D}}(f) = \mathbb{E}_X[\tilde{\mathbf{p}}(X)T^{-1}\vec{\ell}(f(X))]$. Under imperfect estimator $\hat{T}$:

$$\mathbb{E}_{\hat{\mathcal{D}}}[\hat{T}^{-1} \vec{\ell}(f(X))]_{\hat{Y}} = \mathbb{E}_X[\tilde{\mathbf{p}}(X)\hat{T}^{-1} \vec{\ell}(f(X))]. \quad (7)$$

Subtracting clean risk:

$$\left| \mathbb{E}_{\hat{\mathcal{D}}}[\hat{T}^{-1} \vec{\ell}(f(X))]_{\hat{Y}} - R_{\mathcal{D}}(f) \right| = \left| \mathbb{E}_X \left[\tilde{\mathbf{p}}(X)(\hat{T}^{-1} - T^{-1}) \vec{\ell}(f(X)) \right] \right| \quad (8)$$

$$\leq \mathbb{E}_X \left[\|\tilde{\mathbf{p}}(X)\|_2 \|\hat{T}^{-1} - T^{-1}\|_2 \|\vec{\ell}(f(X))\|_2 \right]. \quad (9)$$

Since $\tilde{\mathbf{p}}(X) \in \Delta^{K-1}$, $\|\tilde{\mathbf{p}}(X)\|_2 \leq \|\tilde{\mathbf{p}}(X)\|_1 = 1$. Since each entry of $\vec{\ell}$ is bounded by M , $\|\vec{\ell}(f(X))\|_2 \leq \sqrt{K}M$. Applying Lemma 1:

$$\left| \mathbb{E}_{\hat{\mathcal{D}}}[\hat{T}^{-1} \vec{\ell}(f(X))]_{\hat{Y}} - R_{\mathcal{D}}(f) \right| \leq 1 \cdot \frac{\|T^{-1}\|_2^2 \epsilon}{1 - \|T^{-1}\|_2 \epsilon} \cdot \sqrt{K}M = \frac{\sqrt{K}M \|T^{-1}\|_2^2 \epsilon}{1 - \|T^{-1}\|_2 \epsilon}. \quad (10)$$

□

1.3 Proposition 2B: Finite-Sample Clean Excess Risk Bound

Proposition 2. Assume ℓ is M -bounded and $L_{\ell,2}$ -Lipschitz with respect to the ℓ_2 norm. Let $\hat{T}$ be a fixed matrix or computed from an independent split $S_T \perp S_R$ such that $\|\hat{T} - T\|_F \leq \epsilon < 1/\|T^{-1}\|_2$. Let $\hat{f} = \arg \min_{f \in \mathcal{F}} \hat{R}_{S_R, \hat{T}}(f)$ on independent sample S_R of size n_R . Then with probability at least $1 - \delta$:

$$R_{\mathcal{D}}(\hat{f}) - \min_{f \in \mathcal{F}} R_{\mathcal{D}}(f) \leq \frac{2\sqrt{KM}\|T^{-1}\|_2^2\epsilon}{1 - \|T^{-1}\|_2\epsilon} + 4\sqrt{2}L_{\ell,2}\|\hat{T}^{-1}\|_2\mathcal{R}_{n_R}(\mathcal{F}) + 2\sqrt{KM}\|\hat{T}^{-1}\|_2\sqrt{\frac{\ln(2/\delta)}{2n_R}}. \quad (11)$$

Proof. Let $f^* = \arg \min_{f \in \mathcal{F}} R_{\mathcal{D}}(f)$. By standard algebraic risk decomposition:

$$R_{\mathcal{D}}(\hat{f}) - R_{\mathcal{D}}(f^*) = \left(R_{\mathcal{D}}(\hat{f}) - R_{\tilde{\mathcal{D}}, \hat{T}}(\hat{f})\right) + \left(R_{\tilde{\mathcal{D}}, \hat{T}}(\hat{f}) - \hat{R}_{S_R, \hat{T}}(\hat{f})\right) \quad (12)$$

$$+ \left(\hat{R}_{S_R, \hat{T}}(\hat{f}) - \hat{R}_{S_R, \hat{T}}(f^*)\right) + \left(\hat{R}_{S_R, \hat{T}}(f^*) - R_{\tilde{\mathcal{D}}, \hat{T}}(f^*)\right) \quad (13)$$

$$+ \left(R_{\tilde{\mathcal{D}}, \hat{T}}(f^*) - R_{\mathcal{D}}(f^*)\right). \quad (14)$$

By empirical risk minimisation optimality of $\hat{f}$, $\hat{R}_{S_R, \hat{T}}(\hat{f}) \leq \hat{R}_{S_R, \hat{T}}(f^*)$, so the third term is ≤ 0 . By Proposition 2A, the first and fifth terms satisfy:

$$|R_{\mathcal{D}}(\hat{f}) - R_{\tilde{\mathcal{D}}, \hat{T}}(\hat{f})| \leq \frac{\sqrt{KM}\|T^{-1}\|_2^2\epsilon}{1 - \|T^{-1}\|_2\epsilon}, \quad |R_{\tilde{\mathcal{D}}, \hat{T}}(f^*) - R_{\mathcal{D}}(f^*)| \leq \frac{\sqrt{KM}\|T^{-1}\|_2^2\epsilon}{1 - \|T^{-1}\|_2\epsilon}. \quad (15)$$

The remaining terms are bounded by the uniform generalization deviation:

$$R_{\mathcal{D}}(\hat{f}) - R_{\mathcal{D}}(f^*) \leq \frac{2\sqrt{KM}\|T^{-1}\|_2^2\epsilon}{1 - \|T^{-1}\|_2\epsilon} + 2 \sup_{f \in \mathcal{F}} \left| \hat{R}_{S_R, \hat{T}}(f) - R_{\tilde{\mathcal{D}}, \hat{T}}(f) \right|. \quad (16)$$

Define the function class $\mathcal{G} = \{g_f(x, \tilde{y}) = [\hat{T}^{-1}\vec{\ell}(f(x))]_{\tilde{y}} : f \in \mathcal{F}\}$. For each sample $(x_i, \tilde{y}_i)$, $|g_f(x_i, \tilde{y}_i)| \leq \|\hat{T}^{-1}\|_2\|\vec{\ell}(f(x_i))\|_2 \leq \sqrt{KM}\|\hat{T}^{-1}\|_2$. Applying McDiarmid's inequality to the random variable $\Phi(S_R) = \sup_{f \in \mathcal{F}} |\hat{R}_{S_R, \hat{T}}(f) - R_{\tilde{\mathcal{D}}, \hat{T}}(f)|$ with bounded differences $c_i = \frac{2\sqrt{KM}\|\hat{T}^{-1}\|_2}{n_R}$, with probability at least $1 - \delta$:

$$\sup_{f \in \mathcal{F}} \left| \hat{R}_{S_R, \hat{T}}(f) - R_{\tilde{\mathcal{D}}, \hat{T}}(f) \right| \leq 2\mathcal{R}_{n_R}(\mathcal{G}) + \sqrt{KM}\|\hat{T}^{-1}\|_2\sqrt{\frac{\ln(2/\delta)}{2n_R}}. \quad (17)$$

By Maurer's vector-valued Rademacher contraction inequality for vector Lipschitz functions:

$$\mathcal{R}_{n_R}(\mathcal{G}) \leq \sqrt{2}L_{\ell,2}\|\hat{T}^{-1}\|_2\mathcal{R}_{n_R}(\mathcal{F}). \quad (18)$$

Combining terms gives the stated bound. $\square$

2 Full Statistical Test Matrix

Table 1 provides the complete inferential comparison across all 24 pairs, documenting both unadjusted p -values and family-wise Holm-Bonferroni corrected thresholds.

Table 1: Complete Statistical Significance Matrix: Unadjusted and Holm-Bonferroni Corrected Tests vs CE Baseline across $N = 3$ Seeds ($df = 2$).

Regime	Comparison	Mean Δ Acc (%)	t -statistic	Exact p	Holm p	Cohen's d
Clean	GCE vs CE	-1.80	-22.44	0.0020	0.0442	-12.96
Clean	SCE vs CE	-10.80	-14.31	0.0048	0.0885	-8.26
Clean	Forward (True T) vs CE	-0.18	-3.93	0.0591	0.2955	-2.27
Clean	Forward (Anchor T) vs CE	-0.20	-5.01	0.0375	0.2599	-2.89
Clean	Forward (ConfLearning T) vs CE	-0.63	-3.78	0.0634	0.2955	-2.18
Clean	Forward (Perturbed T) vs CE	-0.91	-3.77	0.0637	0.2955	-2.18
Sym 20%	GCE vs CE	+3.15	+11.81	0.0071	0.1091	+6.82
Sym 20%	SCE vs CE	-6.84	-9.67	0.0105	0.1264	-5.58
Sym 20%	Forward (True T) vs CE	+2.35	+22.60	0.0020	0.0442	+13.05
Sym 20%	Forward (Anchor T) vs CE	+1.59	+8.62	0.0132	0.1403	+4.98
Sym 20%	Forward (ConfLearning T) vs CE	+2.89	+10.83	0.0084	0.1095	+6.25
Sym 20%	Forward (Perturbed T) vs CE	+3.62	+14.60	0.0047	0.0885	+8.43
Sym 50%	GCE vs CE	+7.95	+11.39	0.0076	0.1091	+6.58
Sym 50%	SCE vs CE	-23.42	-17.69	0.0032	0.0636	-10.21
Sym 50%	Forward (True T) vs CE	+4.72	+27.89	0.0013	0.0308	+16.10
Sym 50%	Forward (Anchor T) vs CE	+3.88	+5.04	0.0371	0.2599	+2.91
Sym 50%	Forward (ConfLearning T) vs CE	+5.52	+8.77	0.0128	0.1403	+5.06
Sym 50%	Forward (Perturbed T) vs CE	+7.55	+14.18	0.0049	0.0885	+8.19
Asym 40%	GCE vs CE	-2.88	-8.67	0.0131	0.1403	-5.00
Asym 40%	SCE vs CE	-18.47	-8.77	0.0128	0.1403	-5.06
Asym 40%	Forward (True T) vs CE	+6.19	+22.77	0.0019	0.0442	+13.15
Asym 40%	Forward (Anchor T) vs CE	+2.01	+1.35	0.3087	0.6174	+0.78
Asym 40%	Forward (ConfLearning T) vs CE	+0.97	+0.99	0.4263	0.6174	+0.57
Asym 40%	Forward (Perturbed T) vs CE	+5.42	+12.05	0.0068	0.1091	+6.96